AI & DATA SCIENCE USING PYTHON

Course Highlights

Python: Core to Advanced, OOP, File I/O, SQL, Web with Flask

Data Science: Statistics, Probability, EDA, Visualization, Power BI

Machine Learning: 15+ Algorithms, Preprocessing, Evaluation, Deployment

Deep Learning: ANN, CNN, RNN, LSTM using TensorFlow and Keras

Gen AI and Agentic AI: LLMs, RAG, Vector DBs, LangGraph, CrewAI

Python Programming

Python Overview

- Installation of Anaconda Python,
- Python Features
  - Variables, Operators, Data Types
  - Conditions, Loops
- Functions, Modules, Packages
- String object, Exercises
- List object, Exercises, Case study
- Tuple object, Exercises
- Dictionary object, Exercises, Case Study
- Set, Frozenset
- Comprehensions
  - List comprehension,
  - Dictionary comprehension.
  - Set Comprehension
- Regular expressions
  - Identifiers
  - Quantifiers
  - Exercises
- Python File I/O
  - Applications using File IO
  - Exercises
- Exception Handling

- Generator functions and Generator Expressions
- Decorators

- SQL Database SQLITE
  - CRUD operations, SQL Queries, Kinds of Joins
  - Project on SQL database with Sqlite

- Working with JSON data
- Working with XML data,
- Working with PDF data
- Working with CSV data
- Generators, Iterators,
  - Hacker Rank problem solving,

- **Object Oriented Python,**
  - Class, Object
  - Abstraction
  - Encapsulation
  - Inheritance
  - Polymorphism
    - Case studies

Mathematics for Data Science and ML

Mathematical Computing with Python (NumPy)

- Introduction to NumPy, N-D array
- Data types and attributes of Arrays
- Mathematical Functions of NumPy
- Array Indexing and Slicing
- Array broadcasting
  - Comparing Core Python Objects with Numpy, Exercises
  - Case study

Statistics

- Central Tendency (mean, median and mode)
- Measures of Variation (Interquartile Range, Variance, Standard Deviation)
- Bar Chart, Histogram, Box whisker plot, Scatter Plot

- Co-variance, Correlation
- Central Limit Theorem,
- Skewness and Kurtosis
- Z Test, T Test, P-value,
- Hypothesis testing
- Chi-Square, Sampling Techniques
- ANOVA (Analysis of Variance)

Probability

- Introduction to Probability, Uncertainty, Random numbers
- Joint Probability, Marginal Probability, Conditional Probability, Exclusivity
- Probability Distributions (PMF, CDF, Normal Distribution, ..)
- Bayes Theorem.

DATA ANALYTICS

• Introduction to Pandas

• Understanding Series, DataFrame, Panel

• Transforming List, Tuple, Dictionaries into Data Frame

• Accessing rows and columns, Iteration over Data Frames

- Pandas joining and merging,
- Pandas Groupby, Pivot Table, Binning,
- Pandas Visualization
- Data Generation
    - Real time Case Studies on Data Analysis based on Kaggle

Exploratory Data Analysis and Data Visualization

- Univariant analysis and Bivariant analysis
- Data Visualization
    - MatplotLib data visualization
    - Seaborn data visualization

Power BI

- Business Intelligence (BI)
- Loading data into PowerBI
- Working with Power Query Editor
- Working with Report Section & Visuals in Power BI

Front end and Web Application development

User Interface for Model Deployment

- HTML:- Basic Elements, Lists, Tables, Forms, Examples
- CSS: Syntax, Selectors, inline/internal/external CSS examples,
- Flask Framework: Environment,

  Routing, URL building, HTTP methods, Templates,

  Static files, Request object, Flask with Sqlite database, case study
- Streamlit Framework, Gradio Framework
- FastAPI
  - Pydantic for data validation
- Web scraping using beautifulsoup4, requests libraries
  - Case Studies on web scraping.
  - Git/GitHub version control

Machine Learning

Data Preprocessing Techniques

- Data Imputation (Missing values)
    - Simple Imputation
    - KNN Imputation
    - Iterative Imputation

- Data Encoding Techniques
    - Label Encoding
    - OneHot Encoding

- Finding Outliers
    - IQR technique
    - Zscore technique
    - Percentile technique
    - LocalOutlierFactor

- Data Normalization, Transformation, Scaling
    - MinMaxScaler
    - StandardScaler
    - RobustScaler
    - PowerTransform
    - Box-CoxTransform
    - QuantileTransforms

- Dimensionality Reduction Techniques
    - PCA- Principle Component Analysis
    - SVD- Singular Value Decomposition
    - LDA- Linear Discriminant analysis

- Feature Selection (Importance) and Engendering techniques
    - Supervised learning based
    - Unsupervised learning based
    - feature importance interpretation (SHAP, LIME)

- Case Studies on Data Preprocessing techniques and comparative analysis of various techniques

Regression Analysis

- Regression, Linear regression
- Linear regression, Multiple Regression
- Ridge Regression, Lasso Regression
- Explanation of statistics
- Evaluation metrics (R-Squre, Adj R-Squre, MSE, RMSE)
- Train/Test Split, Hypothesis testing formal way
- Case Studies
- Project on Regression Analysis from Kaggle

Classification

- Introduction to Machine Learning
- Naïve Bayes classifier
- Decision Tree classifier
- KNN classifier
- Logistic Regression
- Support Vector Machines (SVM)
- One-vs-Rest and One-vs-One for Multi-Class Classification
- Predict() Vs PredictProba()
- Ensemble models (Random Forest, Bagging, Boosting)
  - Bagging algorithms
  - Boosting algorithms
  - Stacking algorithms

    Xgboost indepth with Industry cases
- SK Learn ML library using Python and Case Studies
- Project on Classification Algorithms from Kaggle

Model Selection and Evaluation

- Accuracy measurements
- Precision, Recall, Precision – Recall Trade-off
- AUC Score, ROC Curve
- Train/Validation/Test split, K-Fold Cross Validation
- The Problem of Over-fitting (Bias-Variance Trade-off)

Learning Best Practices for Model Evaluation

- Bias, Variance, Overfitting, Underfitting methods
- Pipelining
- Parameter Tuning mechanisms (Grid Search, Random Search)
- Debugging algorithms with learning and validation curves

Case Study

Association Analysis

- Association Rules & Interesting measures
- Apriori Algorithm
- FP-Grouth algorithm
- Case Studiies

Clustering

- Similarity distance measures
- K-means Clustering
- Hierarchical Clustering
- DB Scan Clustering
- Case Studies

Applications of ML

- Time Series Analysis (Stock Market forecasting using ARIMA models)
- Recommendation Systems (Filter based RS and Collaborative based RS)
- Dealing with Imbalanced datasets (Anomaly Detection Methods)

- Deployment ML/DL/NLP models using Flask, Fast API, Github
- ML Model Deployment in Azure

Applications of ML

- Productionizing the Environment
- Experiment Tracking & Data Versioning
- Model Serving & API Development

- CI/CD & Pipeline Automation
- Monitoring & Drift Detection

Natural Language Processing

NLP

- NLP Overview, Applications using NLTK, Text Blob
- Tokenizing, Stop Word Removal, Stemming, Lemmatization, POS Tagging,
- Similarity measures over Text,
- Vector Space Model, Bag of words,
- transforming text to Numeric using Count Vectorizer
- Text Classification,
- Text Clustering,
- Topic Modelling,
- Model Deployment using NLP
- Word Embeddings, Sentiment Analysis
- Case Studies
- Project on NLP from Kaggle

Computer Vision

Image Processing using CV2

- Image Processing Basics and Computer Vision Library
- Images Operations using Numpy
- Edge detection
- Contour detection
- Feature Marching
- Face detection

DEEP LEARNING

Deep Learning using TensorFlow and Kera's.

- Introduction to Deep Learning,
- Neurons, Perceptron, Multilayer Perceptron,
- Forward Propagation, Backward Propagation,
- Activation Functions,
- Gradient Descend Algorithm
- Artificial Neural Networks (ANN)
    - Case Studies
- Convolution Neural Networks (CNN)
    - Case Studies
- Recurrence Neural Networks (RNN)
    - LSTM, GRU algorithms, Case Studies
- Model Deployment using Deep Learning
- Deep Learning with Text
- Word Embedding
- Transformers
    - Encoders and Decoders,
    - Attention mechanism
- Hugging Face Pre-trained models
- **Capstone project on ML/NLP/DL**

GEN AI, RAG Vector Databases, AGENTIC AI

Generative AI & LLMs & LVMs

- Introduction to Gen AI and LLMs
- Architecture Overview
- Limitations and challenges
- Configuration Parameters for LLMs

LLMs:

- Open AI, Gemini, Llama, Mistral, QWEN…
- Working with text, Images, Video
- Creating own Chatbots
- Working wit databases (SQL function calling)
- evaluation metrics for LLMs (BLEU, ROUGE, perplexity)

Prompt Engineering:

- Prompt Engineering Introduction
- Prompt Engineering and Types
- Few-shots, Zero-shot, Chain-of-Thoughts, Instructional Prompting
- Role-Playing and Open-ended Prompting

Ollama :

- Ollama Overview - What is Ollama and Advantages
- Ollama Model Parameters
- Ollama CLI Commands -Pull and Testing a Model
- Summarization and Sentiment Analysis
- Different Ways to Interact with Ollama Models
- Interact with Llama3 in Python Using Ollama REST API

Vector Databases and RAG and CAG

- RAG Introduction
- RAG Key Components,
- Introduction to Vector Databases Overview
- Why Vector Databases, Vector Databases - Benefits and Advantages
- Traditional vs. Vector Databases - Limitations and challenges

- Building Vector Databases
- Chroma Database workflow
- Creating a Chroma DB and Adding Documents and Querying
- Looping Through the Results & Showing Similarity Search Results
- Chroma Vector Database - Persisting Data and Saving
- Creating OpenAI Embeddings - Raw without Chroma
- Using OpenAI Embedding API to Create Embedding in Chroma DB
- Vector Databases Metrics and Data Structures
- Vector Similarity Cosine Similarity
- Euclidean Distance - L2 Norm, Dot Product
- Generating Embeddings from Documents and Insert to Vector Database
- Conversation between Llama/Gemini with Notepad text, PDF, Web data.

Agentic AI Frameworks

- AI Agents, Agentic AI
- AI Agentic Design Patterns

LangChain:

- LangChain Fundamentals
- ChatModel,ChatPromptTemplates
- Indexes, Retrievers and Data Preparation
- TextLoaders
- Text Splitting and Cleaning
- Embeddings and Retriever with FAISS VectorStore
- TextSplitter, DirectoryLoader
- PDFLoader, Chains
- RAG System with Chat and LangChain Chains
- RAG System QA Bot Using LangChain
- Query Expansion with Multiple Queries - Overview
- Query Expansion Full RAG Workflow
- Re-Ranking & Cross-encoder and Bi-encoders - Overview
- Reranking Technique RAG System Workflow Architecture

LangGraph:

- LangGraph Overview, Key Concepts
- Core Concepts - Flow Diagram
- Data and State
- Bot - Streaming Values
- Adding Tools to our Basic LangGraph Agent
- Adding Memory to Our Agent State
- Adding Human-in-the-loop to the AI Agent
- Adding Nodes and Edges and Running our Agent
- Creating REAT agent using LangGraph

CrewAI

- CrewAI Overview
- Configure your LLM backend (OpenAI, Groq, or Ollama).
- Agents & Tools
  - Learn how to define **agents** with:
    - Role, goal, backstory
    - Assign tools (e.g., calculator, web search, custom Python functions).
- Tasks & Crews
  - Learn orchestration models:
    - Sequential (linear steps)
    - Hierarchical (supervisor assigns tasks)
  - Advanced Features
    - **C**ustom tools (e.g., Python functions, database queries).
    - Human-in-the-Loop (review before final output).
    - Implementing RAG CREW

Model Context Protocol:

- MCP overview & history
- Purpose: standardizing model ↔ tool communication
- MCP vs API vs Lang Chain/LLM tool calling
- MCP Architecture
- Build a simple MCP tool (e.g., math_calculator).
- Connect it to a LangChain agent (via custom tool).
- Test LLM → MCP tool call → response.
  - Hands-On case studies

N8N : Automation of Agents

- Automation of Gmail, Google Drive
- Chat with Telegram
- RAG application Automation

- **Capstone project on Gen AI/ Agentic AI**